Quiz 6 — Instructor Key

Friday, October 23, 2026 10 points

1. (7 points)

(a) Power and constant multiple rules:

$$f'(x) = 15x^2.$$

(b) Rewrite $1/x = x^{-1}$ and use the power and difference rules:

$$f'(x) = -x^{-2} - 6 = -\frac{1}{x^2} - 6, \qquad x \neq 0.$$

(c) Chain rule with an outer power:

$$g'(x) = 4(3x+1)^3(3) = 12(3x+1)^3.$$

(d) Product rule and the derivative of e^x:

$$h'(x) = (2x)e^x + (x^2+1)e^x = e^x(x^2+2x+1).$$

(e) Quotient rule:

$$r'(x) = \frac{(2x)(x-1) - (x^2+3)(1)}{(x-1)^2} = \frac{x^2-2x-3}{(x-1)^2}, \qquad x \neq 1.$$

(f) Natural logarithm and chain rule:

$$s'(x) = \frac{1}{2x+1} \cdot 2 = \frac{2}{2x+1}, \qquad x > -\tfrac{1}{2}.$$

(g) Rewrite the square root as a power and use the chain rule:

$$u'(x) = \frac{1}{2}(3x-4)^{-1/2}(3) = \frac{3}{2\sqrt{3x-4}}, \qquad x > \tfrac{4}{3}.$$

2. (3 points) For $y = \ln(x)$, $y' = 1/x$, so the slope at $x = e$ is $1/e$. The tangent line through $(e, 1)$ is

$$y - 1 = \frac{1}{e}(x - e), \qquad \text{or} \qquad y = \frac{x}{e}.$$

Grading: accept equivalent exact forms, including unsimplified product-rule and quotient-rule expressions. Give partial credit for correct rule setup when an arithmetic error follows. Carry a minor error forward without penalizing it again when the subsequent reasoning is correct. The domain notes in 1(b), 1(e)–(g) are for reference; the quiz does not ask students to state domains.